THE FUNDAMENTAL UNIT OF LIFE

Class 9 | Science | Chapter 5

THE CELL

The cell is the structural and functional unit of life -- discovered by Robert Hooke (1665) in a cork slice.

- Robert Brown discovered the nucleus (1831).
- Schleiden & Schwann proposed the Cell Theory: all living things are made of cells.
- Virchow modified it: all cells arise from pre-existing cells.
- Cells vary greatly in shape and size -- from 0.1-0.5 micrometre (bacteria) to as long as a few metres (nerve cell).

PROKARYOTIC VS EUKARYOTIC CELLS

Feature	Prokaryotic	Eukaryotic
Nucleus	No true nucleus (nucleoid)	True, membrane-bound nucleus
Membrane-bound organelles	Absent	Present
Size	0.1-5.0 micrometre	5-100 micrometre
Example	Bacteria, blue-green algae	Plant & animal cells, fungi

CELL ORGANELLES

Structure & Function

DIAGRAM: PLANT VS ANIMAL CELL

PLANT CELL vs ANIMAL CELL (simplified)

CELL MEMBRANE / PLASMA MEMBRANE

Outermost living layer; selectively permeable -- allows movement of some materials in/out. Made of lipids and proteins.

CELL WALL

Found ONLY in plant cells (outside the membrane); made of cellulose; gives shape and rigidity, protects the cell.

NUCLEUS

Control centre of the cell; bound by nuclear membrane; contains chromosomes (DNA + protein) which carry genetic information.

CYTOPLASM

Jelly-like fluid between the nucleus and cell membrane; contains all other organelles; site of many metabolic reactions.

CELL ORGANELLES (CONTD.)

Functions Table

KEY ORGANELLES & THEIR FUNCTIONS

Organelle	Function
Mitochondria	'Powerhouse of the cell' -- releases energy (ATP) via respiration
Endoplasmic Reticulum (ER)	Transport of materials; Rough ER (with ribosomes) makes proteins; Smooth ER makes lipids
Golgi Apparatus	Packaging & dispatch (storage, modification, secretion) of materials made in the ER
Ribosomes	Site of protein synthesis; found on rough ER or free in cytoplasm
Lysosomes	'Suicide bags' -- contain digestive enzymes; break down waste and worn-out cell parts
Plastids (in plants)	Chloroplast (green, photosynthesis), Chromoplast (colour), Leucoplast (storage)
Vacuole	Storage of water/food/waste; large & permanent in plant cells, small/temporary in animal cells

DIFFUSION & OSMOSIS

Diffusion

Movement of particles from a region of higher concentration to lower concentration.

Osmosis

Movement of water molecules from a region of higher water concentration to lower water concentration through a semi-permeable membrane.

Solution Type	Effect on Cell
Hypotonic (lower conc. outside)	Water enters cell -> cell swells (may burst)
Isotonic (equal conc.)	No net movement of water
Hypertonic (higher conc. outside)	Water leaves cell -> cell shrinks (plasmolysis in plants)

SUMMARY & REVISION

Key Terms + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Cell** -> basic structural & functional unit of life
- **Nucleus** -> control centre, holds DNA
- **Mitochondria** -> powerhouse of the cell
- **Cell wall** -> rigid outer layer, plants only
- **Diffusion** -> movement of particles high to low concentration
- **Osmosis** -> movement of water across a semi-permeable membrane

SOLVED EXAMPLE

Q: Why is the mitochondrion called the 'powerhouse of the cell'?

A: Because it breaks down food (respiration) to release energy in the form of ATP, which the cell uses for its activities.

ONE-LINE RECAP FOR REVISION

Cell = structural + functional unit of life (Robert Hooke, 1665).

Plant cells have a cell wall & plastids; animal cells don't.

Mitochondria = energy; Nucleus = control; Ribosomes = protein synthesis.

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